



Mini ERP: From Demand to Delivery

Overview

Modern businesses still struggle with disconnected spreadsheets, manual inventory tracking, delayed procurement decisions, and lack of visibility across Sales, Purchase, Manufacturing, and Stock operations.

Your challenge is to build a **Mini ERP System** that digitally manages the complete business flow:

- Product Management
- Sales Management
- Purchase Management
- Manufacturing
- Bill of Materials (BoM)
- Inventory & Stock Tracking
- Procurement Automation

The goal is to help businesses move from:	
✗ Manual operations	✓ Centralized operations
✗ Spreadsheet dependency	✓ Real-time stock visibility
✗ Stock confusion	✓ Automated procurement
✗ Delayed procurement	✓ Manufacturing tracking
✗ Production bottlenecks	✓ End-to-end traceability

The Business Problem

Meet “Shiv Furniture Works”

Shiv Furniture is a growing furniture manufacturing company.

Initially, they managed operations using:

- Excel sheets
- WhatsApp messages
- Manual stock registers
- Paper-based manufacturing notes

As customer demand increased, problems started appearing:

Sales Team Problems

- Sales team sold products without checking stock.
- Customers received delayed deliveries.
- No visibility on available inventory.

Purchase Team Problems

- Purchase managers didn't know when raw materials were running low.
- Vendors received urgent last-minute purchase requests.

Manufacturing Team Problems

- Operators didn't know what to manufacture next.
- BoMs were maintained on paper.
- Work orders were not tracked.

Inventory Problems

- No accurate stock balance.
- Components consumed manually.
- Finished goods not updated correctly.

Management Problems

Owners had no visibility into:

- Pending orders
 - Production delays
 - Material shortages
 - Manufacturing efficiency
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The Goal

Build a centralized Mini ERP system that allows Shiv Furniture to:

- Manage products and inventory
 - Process sales and purchase operations
 - Manufacture products using BoMs
 - Track stock movement automatically
 - Trigger procurement automatically
 - Support Make To Stock (MTS)
 - Support Make To Order (MTO)
 - Maintain audit logs for traceability
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Core ERP Concept

The entire ERP revolves around one thing: **Inventory Movement**

Every module affects stock:

Module	Impact
Sales	Decreases stock
Purchase	Increases stock
Manufacturing Components	Consumes stock
Manufacturing Finished Goods	Produces stock
Procurement	Replenishes stock

Core Modules

The ERP consists of 5 major modules:

1. Products
2. Sales
3. Purchase
4. Manufacturing
5. Bill of Materials (BoM)

Additional:

1. Audit Logs
 2. User Access Rights
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Target Users

User Type	Responsibility
Admin	Full system access
Sales User	Manage sales orders
Purchase User	Manage purchase orders
Manufacturing User	Handle manufacturing orders
Inventory Manager	Track stock movement
Business Owner	Monitor business flow (manages product)

Authentication & Access Rights

Each module should support role-based access:

Access Type	Description
Admin	Full access
User	Limited module access
None	No access

Example:

- Sales Users may only access Sales.
 - Manufacturing Users may only access Manufacturing.
 - Admin can access everything including Audit Logs.
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Core Business Flow

Step 1 – Product Creation

Products are created with:

- Sales Price
- Cost Price
- Stock Quantity
- Procurement Strategy

Example:

- Wooden Table
 - Office Chair
 - Dining Table
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Step 2 – Procurement Strategy

Every product can use either:

MTS – Make To Stock

OR

MTO – Make To Order

Understanding MTS (Make To Stock)

Products are manufactured or purchased BEFORE customer demand.

Example: Make To Stock

Current Stock:

- Wooden Chair = 100 Units

Customer orders:

- 10 Wooden Chairs

Result

The system delivers directly from stock.

No procurement needed.

Understanding MTO (Make To Order)

Products are manufactured or purchased ONLY when a customer order arrives.

Example: Make To Order

Current Stock:

- Wooden Table = 2 Units

Customer orders:

- 10 Wooden Tables

Shortage:

- 8 Units

System should automatically:

- Create Purchase Order
OR
 - Create Manufacturing Order
- depending on procurement configuration.
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Procurement Automation

Each product can be configured with

Field	Description
Procure on Demand	Enable automatic replenishment
Procurement Type	Purchase / Manufacturing
Vendor	Used for Purchase
BoM	Used for Manufacturing

Example Procurement Flow

Scenario

Customer orders:

- 20 Dining Tables

Available stock:

- 5

Shortage:

- 15

If Procurement Type = Manufacturing:

System auto creates:

- Manufacturing Order for 15 units

If Procurement Type = Purchase:

System auto creates:

- Purchase Order for 15 units
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Product Module

Products act as the central inventory model.

Features

- Product creation
 - Sales price
 - Cost price
 - On-hand quantity
 - Free-to-use quantity
 - Reserved quantity
 - Procurement setup
 - Stock visibility
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Inventory Concepts

On Hand Quantity

Actual physical stock.

Reserved Quantity

Stock already committed to:

- Sales Orders
 - Manufacturing Orders
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Free To Use Quantity

Available stock after reservations.

Formula

Free To Use Qty = On Hand Qty – Reserved Qty

Sales Module

The Sales module manages customer demand.

Features

- Create Sales Orders
 - Select customer
 - Add products
 - Check stock availability
 - Deliver products
 - Trigger procurement
 - Update inventory
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Sales Workflow

Draft → Confirmed → Partially Delivered → Fully Delivered

OR

Draft → Cancelled

Sales Business Logic

When Sales Order is confirmed:

- Product availability is checked.
- Quantity becomes reserved.
- Procurement may trigger automatically.

When delivered:

- Stock decreases.
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Purchase Module

The Purchase module replenishes stock.

Features

- Create Purchase Orders
 - Vendor management
 - Receive products
 - Increase stock automatically
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Purchase Workflow

Draft → Confirmed → Partially Received → Fully Received

Purchase Business Logic

When products are received:

- On Hand Quantity increases.
 - Stock Ledger updates automatically.
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Manufacturing Module

Manufacturing converts raw materials into finished goods.

Understanding Manufacturing

To manufacture products, businesses need:

- Components
 - Operations
 - Work Centers
 - Work Orders
 - BoM
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Bill of Materials (BoM)

A BoM defines:

- Components needed
 - Quantity needed
 - Operations required
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Example BoM

Product

Wooden Table

Components

Component	Qty
Wooden Legs	4
Wooden Top	1
Screws	12

Operations

Operation	Duration
Assembly	60 mins
Painting	30 mins
Packing	20 mins

Manufacturing Order (MO)

An MO is created to manufacture products.

MO Includes

- Finished Product
 - Quantity
 - Components
 - Work Orders
 - Assignee
 - BoM
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Work Orders

Work Orders are individual production steps.

Example:

- Assembly
 - Painting
 - Packing
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Work Centers

Physical locations where operations happen.

Examples:

- Assembly Line
 - Paint Floor
 - Packaging Unit
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Manufacturing Flow

Step 1

Create Manufacturing Order.

Step 2

System fetches:

- Components
- Operations
- Durations

from BoM.

Step 3

Components become reserved.

Step 4

Operators execute Work Orders.

Step 5

Finished goods added to stock.

Step 6

Consumed components deducted from stock.

Stock Ledger Concept

Stock Ledger tracks every inventory movement.

Example – Manufacturing 10 Tables

Inventory Movement Summary

Product	Movement
Legs	-40
Tops	-10
Screws	-120
Tables	+10

Audit Logs

Every important change must be tracked.

Examples:

- Status changes
 - Quantity changes
 - Price updates
 - Deliveries
 - Manufacturing completion
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Dashboard Requirements

The ERP should provide a real-time dashboard with:

- Total Sales Orders
 - Pending Deliveries
 - Manufacturing Orders
 - Delayed Orders
 - Total Purchase Orders
 - Partial Receipts
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Final Objective

Build a modular Mini ERP platform capable of:

- Managing inventory
- Processing sales & purchases
- Manufacturing finished goods
- Automating procurement
- Supporting MTS & MTO flows
- Maintaining stock traceability
- Tracking audit logs
- Providing operational visibility

You're not just building an ERP—you are building the digital backbone of a growing business. Every order, purchase, manufacturing task, and inventory movement must work together as one connected system.

The best solutions won't just manage data—they will orchestrate an entire business.

Now it's your turn to make an IMPACT.

Code smart. Build big. Have fun. 🚀

Mockup Link -

<https://link.excalidraw.com/l/65VNwvy7c4X/gt2p7234Do>